

화학수학

세 번째 수업

다변수 미분

오늘의 수업

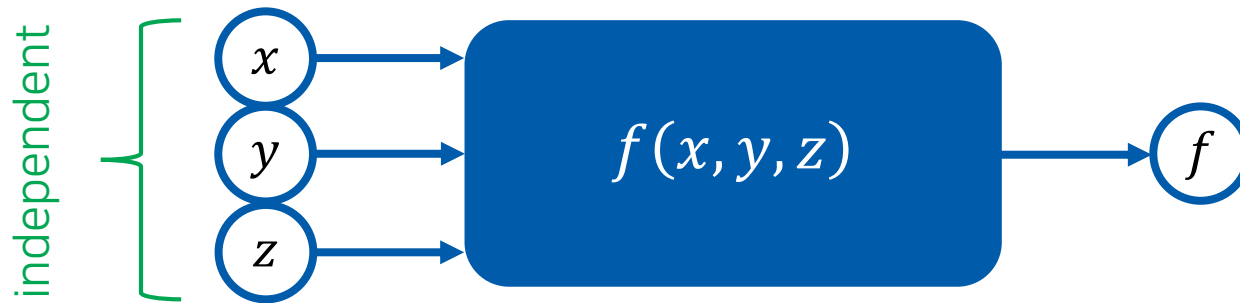
- Functions of several independent variables
- Partial derivatives
 - Fundamental equation of differential calculus
 - Manipulating techniques

8. Differential Calculus with Several Independent Variables

(8장 다변수 미분)

Function of Several Independent Variables

(다변수함수)



Example: $f(x, y, z) = x^2 + xy + 3z$

$$f(1, 1, 1) = 1 + 1 + 3 = 5$$

$$f(-1, 1, 0) = 1 - 1 + 0 = 0$$

⋮

Examples from Physical Chemistry

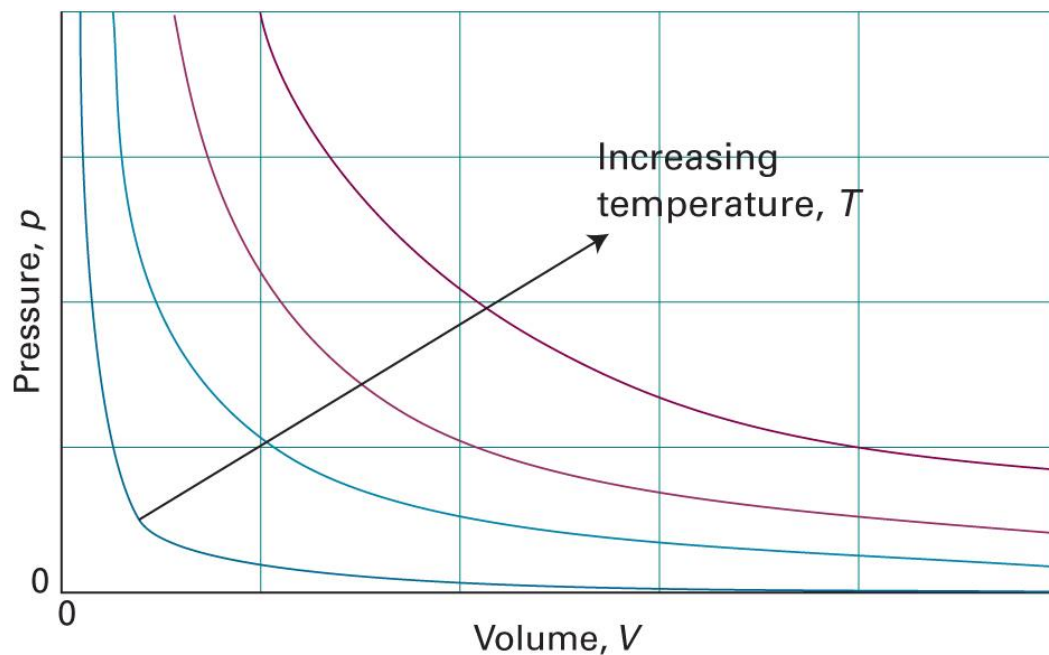
- Thermodynamic properties
 - Pressure $P = P(T, V, n)$
 - Internal energy $U = U(T, V, n)$
- Wave function $\psi = \psi(x, y, z)$
- Reaction rate $r = r([A], [B])$
- piecewise continuous and single-valued

Example: Ideal Gas Equation

$$P = P(T, V, n) = \frac{nRT}{V}$$

Visualization: Single Independent Variable

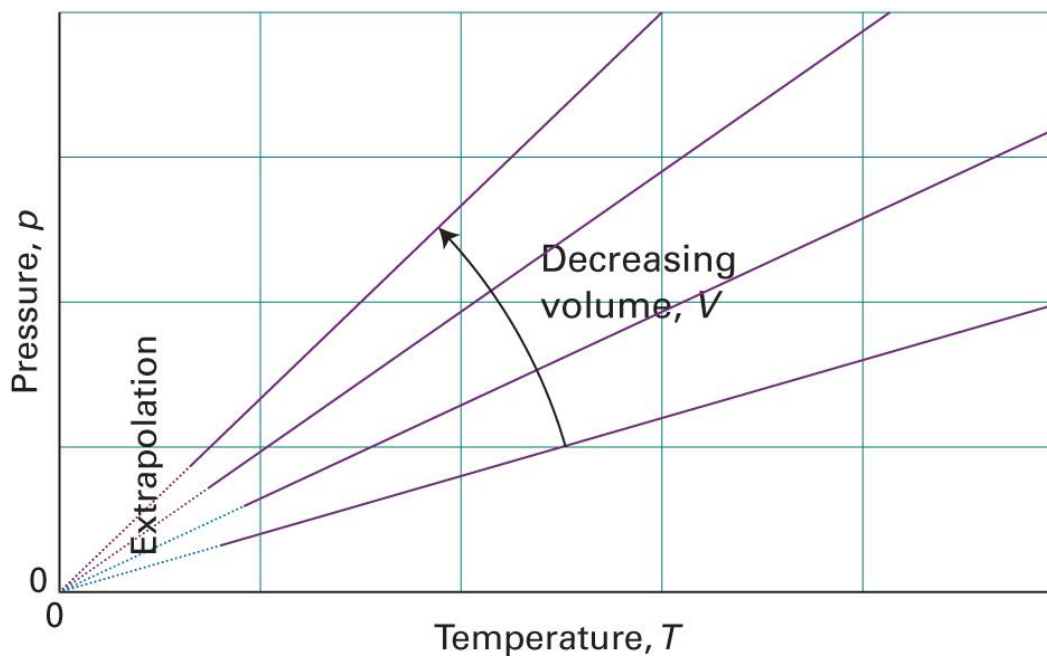
$$P = P(T, V, n) = \frac{nRT}{V}$$



(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 11th ed., Figure 1A.2)

Visualization: Single Independent Variable

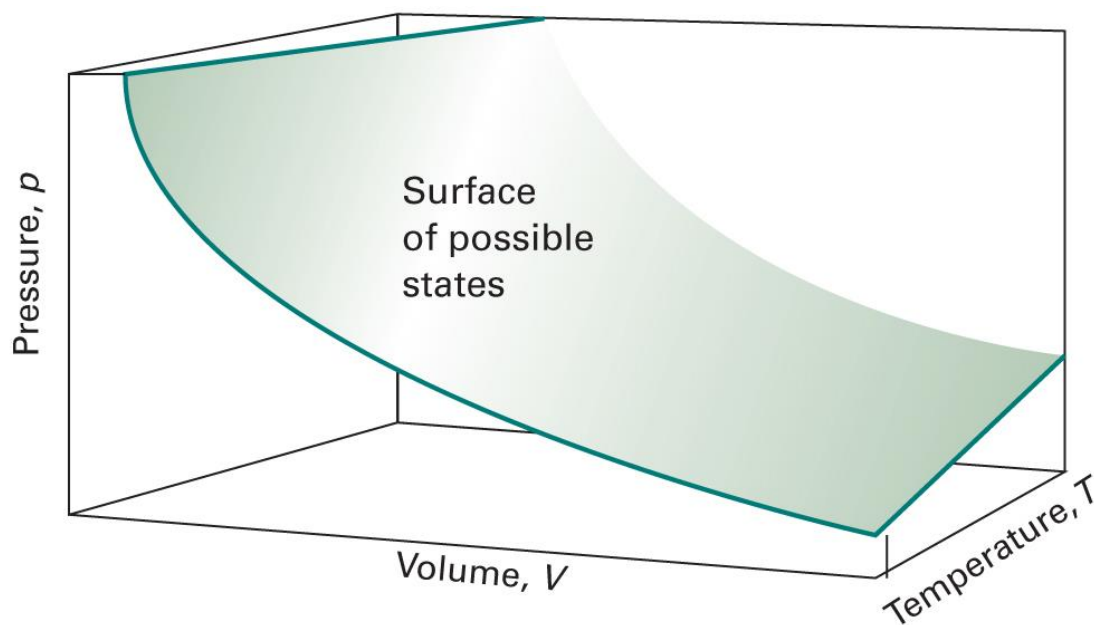
$$P = P(T, V, n) = \frac{nRT}{V}$$



(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 11th ed., Figure 1A.5)

Visualization: Two Independent Variables

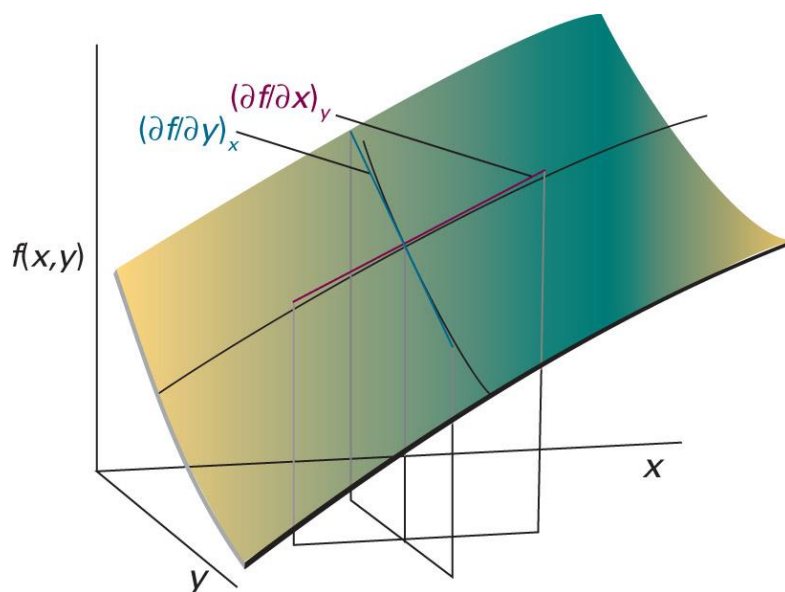
$$P = P(T, V, n) = \frac{nRT}{V}$$



(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 11th ed., Figure 1A.6)

Partial Derivatives (편미분)

“the slope of the function with respect to one of the variables,
all the other variables being held constant”



$$\left(\frac{\partial f}{\partial x}\right)_y = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

$$\left(\frac{\partial f}{\partial y}\right)_x = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

Calculation of Partial Derivatives

Treat fixed variables like **ordinary constants**

$$\left(\frac{\partial P}{\partial V}\right)_{n,T} = \left(\frac{\partial}{\partial V} \left[\frac{nRT}{V} \right]\right)_{n,T} = -\frac{nRT}{V^2}$$

$$\left(\frac{\partial P}{\partial T}\right)_{n,V} = \left(\frac{\partial}{\partial T} \left[\frac{nRT}{V} \right]\right)_{n,V} = \frac{nR}{V}$$

$$\left(\frac{\partial P}{\partial n}\right)_{T,V} = \left(\frac{\partial}{\partial n} \left[\frac{nRT}{V} \right]\right)_{T,V} = \frac{RT}{V}$$

Infinitesimal Changes

For $P = P(T, V, n)$,

$$dP = \left(\frac{\partial P}{\partial V} \right)_{n,T} dV \quad (n \text{ and } T \text{ fixed})$$

$$dP = \left(\frac{\partial P}{\partial V} \right)_{n,T} dV + \left(\frac{\partial P}{\partial T} \right)_{n,V} dT \quad (n \text{ fixed})$$

$$dP = \left(\frac{\partial P}{\partial V} \right)_{n,T} dV + \left(\frac{\partial P}{\partial T} \right)_{n,V} dT + \left(\frac{\partial P}{\partial n} \right)_{T,V} dn$$

↑
(total) differential of P (P 에 대한 전미분)

Fundamental Equation of Differential Calculus

(미분학의 기본 방정식)

- n independent variables: $x_1, x_2, \dots, x_n$
- dependent variable: y

The differential of y is

$$dy = \sum_{i=1}^n \left(\frac{\partial y}{\partial x_i} \right)_{x'} dx_i$$

where x' means keeping all of the variables except for x_i fixed

Case of Ideal Gas

$$dP = \left(\frac{\partial P}{\partial V} \right)_{n,T} dV + \left(\frac{\partial P}{\partial T} \right)_{n,V} dT + \left(\frac{\partial P}{\partial n} \right)_{T,V} dn$$

$$dP = -\frac{nRT}{V^2} dV + \frac{nR}{V} dT + \frac{RT}{V} dn$$

Approximately,

$$\Delta P \approx -\frac{nRT}{V^2} \Delta V + \frac{nR}{V} \Delta T + \frac{RT}{V} \Delta n$$

Example 8.1

Calculate the approximate change in pressure of an ideal gas if

- the volume is changed from 20.000 L to 19.800 L,
- the temperature is changed from 298.15 K to 299.00 K, and
- the amount of gas in moles is changed from 1.0000 mol to 1.0015 mol.

Compare the result with the correct value of the pressure change.

$$\Delta P \approx -\frac{nRT}{V^2} \Delta V + \frac{nR}{V} \Delta T + \frac{RT}{V} \Delta n$$

In SI units, $R = 8.3145 \text{ J K}^{-1} \text{ mol}^{-1}$, and

$V = 0.020000 \text{ m}^3$, $T = 298.15 \text{ K}$, $n = 1.0000 \text{ mol}$ (initial values);

$\Delta V = -2.00 \times 10^{-4} \text{ m}^3$, $\Delta T = 0.85 \text{ K}$, $\Delta n = 1.5 \times 10^{-3} \text{ mol}$



Substitution leads to $\Delta P = 1.779 \times 10^3 \text{ N m}^{-2} = 1.779 \times 10^3 \text{ Pa}$

Example 8.1

Calculate the approximate change in pressure of an ideal gas if

- the volume is changed from 20.000 L to 19.800 L,
- the temperature is changed from 298.15 K to 299.00 K, and
- the amount of gas in moles is changed from 1.0000 mol to 1.0015 mol.

Compare the result with the correct value of the pressure change.

Approximate value: $\Delta P = 1.779 \times 10^3 \text{ N m}^{-2} = 1.779 \times 10^3 \text{ Pa}$

Correct value:

$$\Delta P = P_2 - P_1 = \frac{n_2 R T_2}{V_2} - \frac{n_1 R T_1}{V_1}$$

Substitution leads to $\Delta P = 1.797 \times 10^3 \text{ Pa}$. The error is about 1%.

In General

For $U = U(P, T, n)$,

$$dU = \left(\frac{\partial U}{\partial P} \right)_{T,n} dP + \left(\frac{\partial U}{\partial T} \right)_{P,n} dT + \left(\frac{\partial U}{\partial n} \right)_{P,T} dn$$

Approximately,

$$\Delta U \approx \left(\frac{\partial U}{\partial P} \right)_{T,n} \Delta P + \left(\frac{\partial U}{\partial T} \right)_{P,n} \Delta T + \left(\frac{\partial U}{\partial n} \right)_{P,T} \Delta n$$

Example 8.2

For a sample of 1.000 mol of liquid CCl_4 at a temperature of 20°C and a pressure of 1.000 atm, the following experimental values are available:

$$(\partial U/\partial T)_{P,n} = 129.4 \text{ J K}^{-1}, (\partial U/\partial P)_{T,n} = 8.51 \times 10^{-4} \text{ J atm}^{-1}$$

Estimate the change in the energy of 1.000 mol of CCl_4 if its temperature is changed from 20.0 to 40.0°C and its pressure from 1.0 to 100.0 atm.

Since n is fixed,

$$\Delta U \approx \left(\frac{\partial U}{\partial P}\right)_{T,n} \Delta P + \left(\frac{\partial U}{\partial T}\right)_{P,n} \Delta T + \left(\frac{\partial U}{\partial n}\right)_{P,T} \Delta n = \left(\frac{\partial U}{\partial P}\right)_{T,n} \Delta P + \left(\frac{\partial U}{\partial T}\right)_{P,n} \Delta T$$

From the given values, $\Delta T = 20.0 \text{ K}$ and $\Delta P = 99.0 \text{ atm}$

Substitution leads to $\Delta U = 2588 \text{ J} + 0.084 \text{ J} = 2588 \text{ J}$

Change of Variables

We can use different sets of independent variables:

- $U = U(T, V, n)$

$$dU = \left(\frac{\partial U}{\partial T} \right)_{V,n} dT + \left(\frac{\partial U}{\partial V} \right)_{T,n} dV + \left(\frac{\partial U}{\partial n} \right)_{T,V} dn$$

- $U = U(T, P, n)$

$$dU = \left(\frac{\partial U}{\partial T} \right)_{P,n} dT + \left(\frac{\partial U}{\partial P} \right)_{T,n} dP + \left(\frac{\partial U}{\partial n} \right)_{T,P} dn$$

Be Careful

For most systems,

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} \neq \left(\frac{\partial U}{\partial T}\right)_{P,n}$$

Fixed variables (subscripts) are important!

Example 8.3

Express the function $z = z(x, y) = ax^2 + bxy + cy^2$ in terms of x and u , where $u = xy$. Find the two partial derivatives $(\partial z / \partial x)_y$ and $(\partial z / \partial x)_u$.

As $y = u/x$, substitution leads to a change of variables:

$$z(x, u) = ax^2 + bx \left(\frac{u}{x}\right) + c \left(\frac{u}{x}\right)^2 = ax^2 + bu + \frac{cu^2}{x^2}$$

Now, the partial derivatives are

$$\begin{aligned} \left(\frac{\partial z}{\partial x}\right)_y &= 2ax + by \\ \left(\frac{\partial z}{\partial x}\right)_u &= 2ax - \frac{2cu^2}{x^3} = 2ax - \frac{2cy^2}{x} \end{aligned}$$

different!

Relating Different Sets of Subscripts

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} = \left(\frac{\partial U}{\partial T}\right)_{P,n} + ?$$

Start from one set of independent variables:

$$U = U(T, P, n)$$

$$dU = \left(\frac{\partial U}{\partial T}\right)_{P,n} dT + \left(\frac{\partial U}{\partial P}\right)_{T,n} dP + \left(\frac{\partial U}{\partial n}\right)_{T,P} dn$$

Relating Different Sets of Subscripts

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} = \left(\frac{\partial U}{\partial T}\right)_{P,n} + ?$$

$$dU = \left(\frac{\partial U}{\partial T}\right)_{P,n} dT + \left(\frac{\partial U}{\partial P}\right)_{T,n} dP + \left(\frac{\partial U}{\partial n}\right)_{T,P} dn$$

Divide both sides by dT :

$$\frac{dU}{dT} = \left(\frac{\partial U}{\partial T}\right)_{P,n} \frac{\cancel{dT}}{1} + \left(\frac{\partial U}{\partial P}\right)_{T,n} \frac{dP}{dT} + \left(\frac{\partial U}{\partial n}\right)_{T,P} \frac{dn}{dT}$$

Relating Different Sets of Subscripts

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} = \left(\frac{\partial U}{\partial T}\right)_{P,n} + ?$$

$$\frac{dU}{dT} = \left(\frac{\partial U}{\partial T}\right)_{P,n} + \left(\frac{\partial U}{\partial P}\right)_{T,n} \frac{dP}{dT} + \left(\frac{\partial U}{\partial n}\right)_{T,P} \frac{dn}{dT}$$

Apply the constraints:

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} = \left(\frac{\partial U}{\partial T}\right)_{P,n} + \left(\frac{\partial U}{\partial P}\right)_{T,n} \left(\frac{\partial P}{\partial T}\right)_{V,n} + \left(\frac{\partial U}{\partial n}\right)_{T,P} \cancel{\left(\frac{\partial n}{\partial T}\right)_{V,n}}$$

0

Variable-Change Identity

(변수-변화 항등식)

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} = \left(\frac{\partial U}{\partial T}\right)_{P,n} + \left(\frac{\partial U}{\partial P}\right)_{T,n} \left(\frac{\partial P}{\partial T}\right)_{V,n}$$

Example 8.4

Apply the foregoing method to the function in Example 8.3 and find the relation between $(\partial z / \partial x)_u$ and $(\partial z / \partial x)_y$.

$$\left(\frac{\partial z}{\partial x}\right)_u = \left(\frac{\partial z}{\partial x}\right)_y + ?$$

For $z = z(x, y)$,

$$dz = \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy$$

$$\frac{dz}{dx} = \left(\frac{\partial z}{\partial x}\right)_y + \left(\frac{\partial z}{\partial y}\right)_x \frac{dy}{dx}$$

$$\left(\frac{\partial z}{\partial x}\right)_u = \left(\frac{\partial z}{\partial x}\right)_y + \left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial y}{\partial x}\right)_u$$

Example 8.4

Apply the foregoing method to the function in Example 8.3 and find the relation between $(\partial z / \partial x)_u$ and $(\partial z / \partial x)_y$.

$$\left(\frac{\partial z}{\partial x}\right)_u = \left(\frac{\partial z}{\partial x}\right)_y + \left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial y}{\partial x}\right)_u$$

The function in Example 8.3 is $z = z(x, y) = ax^2 + bxy + cy^2$ and $u = xy$.

Hence,

$$\left(\frac{\partial z}{\partial y}\right)_x = bx + 2cy$$

$$\left(\frac{\partial y}{\partial x}\right)_u = \left(\frac{\partial}{\partial x} \left[\frac{u}{x}\right]\right)_u = -\frac{u}{x^2} = -\frac{y}{x}$$

Example 8.4

Apply the foregoing method to the function in Example 8.3 and find the relation between $(\partial z/\partial x)_u$ and $(\partial z/\partial x)_y$.

$$\left(\frac{\partial z}{\partial x}\right)_u = \left(\frac{\partial z}{\partial x}\right)_y + \left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial y}{\partial x}\right)_u$$

$$\left(\frac{\partial z}{\partial x}\right)_u = \left(\frac{\partial z}{\partial x}\right)_y + (bx + 2cy) \left(-\frac{y}{x}\right)$$

$$\left(\frac{\partial z}{\partial x}\right)_u = \left(\frac{\partial z}{\partial x}\right)_y - by - \frac{2cy^2}{x}$$

Reciprocal Identity

(역 항등식)

$$\left(\frac{\partial y}{\partial x}\right)_{z,u} = \frac{1}{(\partial x / \partial y)_{z,u}}$$

Example 8.5

Show that

$$\left(\frac{\partial P}{\partial V}\right)_{n,T} = \frac{1}{(\partial V / \partial P)_{n,T}}$$

for an ideal gas.

Since $P = nRT/V$,

$$\left(\frac{\partial P}{\partial V}\right)_{n,T} = -\frac{nRT}{V^2}$$

Since $V = nRT/P$,

$$\left(\frac{\partial V}{\partial P}\right)_{n,T} = -\frac{nRT}{P^2} = -nRT \left(\frac{V}{nRT}\right)^2 = -\frac{V^2}{nRT}$$

Second Partial Derivatives

(2계 편도함수)

For $z = z(x, y)$,

$$\left(\frac{\partial^2 z}{\partial x^2}\right)_y = \left[\frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x}\right)_y\right]_y$$

$$\frac{\partial^2 z}{\partial y \partial x} = \left[\frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x}\right)_y\right]_x$$

(Mixed second partial derivative; 혼합 2계 편도함수)

Euler Reciprocity Relation

(오일러 역관계식)

For $z = z(x, y)$,

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$$

Example 8.6

Show that

$$\left(\frac{\partial^2 P}{\partial V \partial T}\right)_n = \left(\frac{\partial^2 P}{\partial T \partial V}\right)_n$$

for an ideal gas.

Since $P = nRT/V$,

$$\left(\frac{\partial^2 P}{\partial V \partial T}\right)_n = \left[\frac{\partial}{\partial V} \left(\frac{nR}{V}\right)\right]_{n,T} = -\frac{nR}{V^2}$$

and

$$\left(\frac{\partial^2 P}{\partial T \partial V}\right)_n = \left[\frac{\partial}{\partial T} \left(-\frac{nRT}{V^2}\right)\right]_{n,V} = -\frac{nR}{V^2}$$

Thermodynamic Relation

$$dU = TdS - PdV + \mu dn$$

For $U = U(S, V, n)$,

$$dU = \left(\frac{\partial U}{\partial S}\right)_{V,n} dS + \left(\frac{\partial U}{\partial V}\right)_{S,n} dV + \left(\frac{\partial U}{\partial n}\right)_{S,V} dn$$

Comparison leads to

$$T = \left(\frac{\partial U}{\partial S}\right)_{V,n}, \quad P = -\left(\frac{\partial U}{\partial V}\right)_{S,n}, \quad \mu = \left(\frac{\partial U}{\partial n}\right)_{S,V}$$

Maxwell Relations

$$T = \left(\frac{\partial U}{\partial S} \right)_{V,n}, \quad P = - \left(\frac{\partial U}{\partial V} \right)_{S,n}, \quad \mu = \left(\frac{\partial U}{\partial n} \right)_{S,V}$$

Since $(\partial^2 U / \partial V \partial S)_n = (\partial^2 U / \partial S \partial V)_n$,

$$\left(\frac{\partial T}{\partial V} \right)_{S,n} = - \left(\frac{\partial P}{\partial S} \right)_{V,n}$$

Relation between apparently unrelated quantities

Cycle Rule

(순환 법칙)

$$\left(\frac{\partial y}{\partial x}\right)_z \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x = -1$$

Start from $y = y(x, z)$;

$$dy = \left(\frac{\partial y}{\partial x}\right)_z dx + \left(\frac{\partial y}{\partial z}\right)_x dz$$

$$\frac{dy}{dx} = \left(\frac{\partial y}{\partial x}\right)_z \frac{dx}{dx} + \left(\frac{\partial y}{\partial z}\right)_x \frac{dz}{dx}$$

1

Cycle Rule

(순환 법칙)

$$\left(\frac{\partial y}{\partial x}\right)_z \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x = -1$$

$$\frac{dy}{dx} = \left(\frac{\partial y}{\partial x}\right)_z + \left(\frac{\partial y}{\partial z}\right)_x \frac{dz}{dx}$$

Fix y ;

$$\left(\frac{\cancel{\partial y}}{\partial x}\right)_y = \left(\frac{\partial y}{\partial x}\right)_z + \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y$$

0

Cycle Rule

(순환 법칙)

$$\left(\frac{\partial y}{\partial x}\right)_z \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x = -1$$

$$0 = \left(\frac{\partial y}{\partial x}\right)_z + \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y$$

$$-1 = \frac{\left(\frac{\partial y}{\partial x}\right)_z}{\left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y} = \left(\frac{\partial y}{\partial x}\right)_z \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x$$

Example 8.8

For the function $y = z \ln x$, show that the cycle rule is valid.

Since $x = e^{y/z}$ and $z = y/(\ln x)$,

$$\left(\frac{\partial y}{\partial x}\right)_z = \frac{z}{x}, \quad \left(\frac{\partial x}{\partial z}\right)_y = -\frac{y}{z^2} e^{y/z}, \quad \left(\frac{\partial z}{\partial y}\right)_x = \frac{1}{\ln x}$$

Hence,

$$\begin{aligned} \left(\frac{\partial y}{\partial x}\right)_z \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x &= \frac{z}{x} \times \left(-\frac{y}{z^2} e^{y/z}\right) \times \frac{1}{\ln x} \\ &= \frac{z}{x} \times \left(-\frac{y}{z^2} x\right) \times \frac{z}{y} = -1 \end{aligned}$$

Chain Rule

(연쇄 법칙)

For $z = z(u, x, y)$ and $x = x(u, v, y)$,

$$\left(\frac{\partial z}{\partial y}\right)_{u,v} = \left(\frac{\partial z}{\partial x}\right)_{u,v} \left(\frac{\partial x}{\partial y}\right)_{u,v}$$

Note:

$$\left(\frac{\partial z}{\partial x}\right)_{u,v} \left(\frac{\partial x}{\partial y}\right)_{u,v} \left(\frac{\partial y}{\partial z}\right)_{u,v} = 1$$

Check the subscripts!

Example 8.9

Show that if $z = ax^2 + bvx$ and $x = uy$, then the chain rule is valid.

Recall the chain rule:

$$\left(\frac{\partial z}{\partial y}\right)_{u,v} = \left(\frac{\partial z}{\partial x}\right)_{u,v} \left(\frac{\partial x}{\partial y}\right)_{u,v}$$

Since $z = z(y, u, v) = a(uy)^2 + bv(uy) = au^2y^2 + buvy$,

$$\left(\frac{\partial z}{\partial y}\right)_{u,v} = 2au^2y + buv$$

$$\left(\frac{\partial z}{\partial x}\right)_{u,v} = 2ax + bv, \quad \left(\frac{\partial x}{\partial y}\right)_{u,v} = u$$

$$\left(\frac{\partial z}{\partial x}\right)_{u,v} \left(\frac{\partial x}{\partial y}\right)_{u,v} = (2ax + bv) \times u = 2au^2y + buv$$

same!

